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**Submission date:** 08-May-2025 10:00AM (UTC+0530)
**Submission ID:** 2355034465
**File name:** ML_Project_Report.pdf.pdf (637.42K)
**Word count:** 4547
**Character count:** 27578

# MACHINE LEARNING PROJECT REPORT

## Hybrid Document Search Engine Using TF-IDF and BERT


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May 2025

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# Candidates Declaration

We, the undersigned, hereby declare that this project report entitled "Hybrid Document Search Engine Using TF-IDF and BERT," submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in the School of Engineering and Technology at BML Munjal University, represents original work carried out by us during the period from February 2025 to May 2025.

We certify that the work presented in this report is authentic and has been conducted under the supervision of Dr. Shilpa Mahajan. All sources of information and assistance received during the project have been duly acknowledged.

Student's Name: Astitva, Vanshit, Vaibhav, Swati, Krish
Date: 6th May, 2025

# Supervisors Declaration

This is to certify that the project report entitled “ Hybrid Document Search Engine Using TF-IDF and BERT" submitted by the above students is a record of the candidates' own work carried out under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree. Faculty Supervisor: Dr. Shilpa Mahajan Signature: Date: 6th May, 2025

# ACKNOWLEDGEMENT

We wish to express our sincere gratitude to Dr. Shilpa Mahajan, Assistant Professor at BML Munjal University, Gurugram, for her exceptional guidance and mentorship throughout our project conducted from February to May 2025. Her expertise, insightful feedback, and constant encouragement were instrumental in navigating the challenges we encountered during our research.

We are deeply thankful to the faculty and administration of the School of Engineering and Technology at BML Munjal University for providing the academic environment and resources that facilitated our work. The knowledge gained through our coursework served as a strong foundation for this project.

Our appreciation extends to our classmates who contributed through collaborative discussions and technical suggestions. Their perspectives enriched our understanding and approach to the problem at hand.

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# ABSTRACT

This project presents a hybrid search engine that combines Term Frequency-Inverse Document Frequency (TF-IDF) with Bidirectional Encoder Representations from Transformers (BERT) to improve document retrieval precision. The goal is to overcome the limitations of keyword-based search (like TF-IDF) by adding semantic understanding through BERT. TF-IDF is used for fast, lexical filtering of documents based on keyword overlap, while BERT computes deep semantic embeddings to accurately rank the most relevant documents. This two-stage pipeline ensures both efficiency and semantic accuracy. The system is particularly useful for academic, legal, and recommendation domains where contextual understanding is vital.

# 1. INTRODUCTION

In the era of digital information overload, retrieving the most relevant document quickly and accurately has become a major challenge across industries like academia, law, healthcare, and e-commerce. As the number of documents grows exponentially, traditional keyword-based search methods fall short in understanding the true intent behind user queries.

Term Frequency-Inverse Document Frequency (TF-IDF) has been one of the most widely used techniques in information retrieval. It calculates the importance of a word in a document relative to a corpus and helps rank documents based on keyword overlap. While TF-IDF is fast and interpretable, it cannot capture contextual meaning, distinguish between polysemous words, or identify synonymous phrases.
With the emergence of transformer-based models like BERT (Bidirectional Encoder Representations from Transformers), semantic search has become significantly more accurate. BERT understands the context and relationships between words, making it capable of identifying relevant documents even if they don't contain the exact query terms. However, its high computational cost makes it impractical for large-scale, real-time document retrieval without pre-filtering.

This project proposes a hybrid document search system that combines the strengths of both TF-IDF and BERT. The system first uses TF-IDF to filter and shortlist a set of documents that are likely relevant based on keyword similarity. BERT is then applied to this smaller set to compute semantic embeddings and rank documents based on their contextual similarity to the query. This two-step approach offers a balance between speed and semantic accuracy.

The system is designed to work with documents in various formats, including PDFs and text files. It supports applications in research, legal search, and content recommendation—anywhere fast and context-aware document selection is required. By narrowing the search space early and leveraging deep learning for final ranking, this hybrid model addresses the limitations of traditional retrieval systems without incurring the full computational cost of applying BERT to an entire corpus.

## 2. Literature Review

| Citation | Research Question/Problem | Methodology | Key Findings |
|---|---|---|---|
| Salton & Buckley, 1988, Information Processing & Management | How do different term weighting schemes affect retrieval performance? | Comparative analysis of TF-IDF variants. | TF-IDF improves retrieval compared to simple term frequency. |
| Devlin et al., 2018, arXiv preprint | How can transformers achieve state-of-the-art NLP results? | Introduced BERT with Masked Language Model. | Rich contextual embeddings for semantic similarity tasks. |
| Nogueira & Lin, 2019, arXiv preprint | Can BERT improve passage re-ranking? | Used BERT for semantic similarity scoring. | Improved re-ranking accuracy vs lexical matching. |
| Gao et al., 2020, arXiv preprint | How to combine sparse and dense IR representations? | Combined exact label matching with dense embeddings. | Simple combinations achieve strong retrieval performance. |
| Thakur et al., 2021, arXiv preprint | How do IR models perform in zero-shot settings? | Created BEIR benchmark for evaluation. | Evaluated IR models across diverse tasks. |

The increasing volume of digital information has driven the need for effective and efficient information retrieval (IR) systems. Traditional keyword-based methods, while fast, often struggle with the semantic gap, failing to retrieve relevant documents that do not share exact keywords with the query. Conversely, more recent semantic search approaches, leveraging techniques like Transformer models, excel at understanding the meaning of text but can be computationally expensive, especially for large document collections. This literature review examines the foundational concepts of TF-IDF and BERT in the context of information retrieval, and then explores the rationale and existing work on hybrid models that combine their strengths.

1. **TF-IDF for Information Retrieval:**
Term Frequency-Inverse Document Frequency (TF-IDF) has been a cornerstone of IR for decades (Salton & Buckley, 1988). It operates on the principle that the importance of a term in a document is proportional to its frequency in the document but inversely proportional to its frequency across the entire document collection. TF-IDF effectively identifies terms that are distinctive to specific documents. Its strengths lie in its simplicity, computational efficiency, and ability to perform well when queries and relevant documents share significant keyword overlap (Ramos, 2003). However, TF-IDF suffers from limitations in handling synonymy, polysemy, and the overall semantic understanding of text (Deerwester et al., 1990).

2. **BERT and Semantic Search:**
Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018) revolutionized Natural Language Processing (NLP) with its ability to learn deep contextualized word embeddings. By pre-training on massive text corpora, BERT captures rich semantic relationships between words and sentences. In information retrieval, BERT and its variants have shown significant improvements in retrieving semantically relevant documents, even when keyword overlap is minimal (Nogueira & Lin, 2019). The ability of BERT to understand the intent behind a query and the meaning of documents makes it a powerful tool for semantic search. However, the computational cost of encoding large document collections and performing real-time inference can be a significant bottleneck.

3. **Hybrid Search Models Combining TF-IDF and BERT:**
To leverage the speed of traditional methods and the semantic understanding of deep learning models, researchers have explored hybrid search architectures. The combination of TF-IDF and BERT aims to create a system that is both efficient and accurate.

A common approach involves using TF-IDF (or similar fast methods like BM25) to perform an initial retrieval of a larger set of candidate documents, which are then re-ranked using BERT based on their semantic similarity to the query (Gao et al., 2020). This two-stage approach reduces the computational burden of applying the expensive BERT model to the entire document collection. The rationale is that TF-IDF can quickly identify documents with some topical relevance based on keywords, and BERT can then refine this set by considering the semantic nuances.

4. **Rationale for Combining TF-IDF and BERT:**

The synergy between TF-IDF and BERT stems from their complementary strengths:

Efficiency: TF-IDF provides a fast and scalable way to narrow down the search space to a potentially relevant subset of documents.

Semantic Understanding: BERT can then perform a more nuanced ranking of these candidates based on semantic similarity, capturing meaning beyond keyword matching.

Balancing Speed and Accuracy: By strategically combining these methods, hybrid models aim to achieve a better balance between retrieval speed and accuracy compared to using either method alone.

5. **Challenges and Future Directions:**

While promising, hybrid models also present challenges:

Determining the Optimal Filtering Threshold: Deciding how many documents to retrieve with TF-IDF before re-ranking with BERT is crucial and can impact both efficiency and recall.

Weighting the Scores: Effectively combining the TF-IDF scores and BERT similarity scores requires careful tuning and may depend on the specific task and data.

Computational Cost of BERT: Even when applied to a filtered set, the encoding and comparison with BERT embeddings can still be computationally intensive for very large collections or real-time applications.

Future research in this area may focus on:

More intelligent filtering mechanisms that can better identify the most promising candidates for semantic re-ranking.

Dynamic weighting strategies that adapt the influence of TF-IDF and BERT based on the query or the characteristics of the retrieved documents.

Exploring more efficient Transformer architectures or distillation techniques to reduce the computational cost of the semantic ranking component.

End-to-end trainable hybrid models that can learn the optimal combination of sparse and dense representations.

# 3. Dataset Description

## 3.1 Dataset Overview

The dataset used in this project comprises a curated collection of text documents in various formats, including PDFs and plain text files. These documents span multiple domains such as education, law, technology, and general articles, representing a realistic corpus for evaluating semantic search systems. The dataset was assembled from publicly available resources and internal academic content. Each file serves as an individual document and contains unstructured text of varying lengths, styles, and complexity. The goal of this dataset is to simulate real-world document retrieval challenges, where documents may use different terminologies or phrasing to convey similar information.

All files were programmatically loaded and parsed using a custom pipeline that supports format detection, encoding resolution, and text extraction. This approach ensures robustness in dealing with noisy or inconsistently formatted data.

## 3.2 Key Variables

| Attribute | Type | Description |
|---|---|---|
| file_name | Text | Document filename |
| file_format | Text | Type of file (.pdf, .txt) |
| raw_text | Text | Extracted full text |
| doc_id | Integer | Unique document ID |
| preprocessed_text | Text | Cleaned and normalized text |
| tfidf_vector | Sparse Vector | TF-IDF vector for keyword search |
| bert_embedding | Dense Vector | BERT-based vector for semantic sim |

### 3.3 Data Quality and Preprocessing

- File Format Handling: The dataset includes a mix of .txt and .pdf files. For PDFs, the PyPDF2 library was used to extract text. Encoding detection was performed using the chardet library for non-PDF files to handle character encodings like UTF-8, Latin-1, etc.
- Missing or Corrupt Documents: Files that were empty, unreadable, or failed extraction were skipped and logged. Approximately 95% of files were successfully parsed.
- Text Normalization: All documents underwent normalization steps including lowercasing, removal of special characters, extra whitespace trimming, and optional stopword filtering (for TF-IDF).
- Tokenization: Tokenization was performed using the tokenizer associated with the BERT model (bert-base-uncased). This ensures compatibility during the semantic embedding phase.
- Vector Representations:
  - TF-IDF: A TfidfVectorizer was trained on the full corpus to convert preprocessed text into sparse numerical vectors for initial filtering.
  - BERT Embeddings: Documents and queries were encoded using BERT into 768-dimensional embeddings with mean pooling for ranking.
- Storage: The TF-IDF matrix and vectorizer were saved using pickle for reuse, while BERT embeddings were computed on-the-fly for filtered documents to save memory.

# 4. Methodology

This section outlines the complete pipeline developed for the hybrid document search system. Our methodology comprises four primary stages: data preprocessing, semantic vector representation, model training and evaluation, and performance assessment.

## 4.1 Data Preprocessing

The data preprocessing stage was essential to ensure that the documents were properly prepared for vectorization and semantic encoding. The system was designed to support both PDF and plain text (.txt) files. For PDF files, textual content was extracted using the PyPDF2 library. Non-PDF documents were read using chardet-based encoding detection to accommodate various character encodings such as UTF-8 and ISO-8859-1.

Once the raw text was obtained, standard text normalization was performed. This included lowercasing, removal of extra whitespaces, and elimination of non-alphanumeric characters. Documents with insufficient textual content were filtered out. Tokenization was performed later during TF-IDF vectorization and BERT encoding using their respective tokenizers.

## 4.2 Semantic Vector Representation

To support both lexical and contextual understanding of text, we implemented two parallel vectorization strategies.

- TF-IDF Vectorization: Using scikit-learn's TfidfVectorizer, we transformed the preprocessed documents into sparse vectors. This allowed us to capture term importance and perform fast keyword-based filtering using cosine similarity.
- BERT Embeddings: For semantic encoding, we used a pre-trained BERT model (bert-base-uncased) from the Hugging Face Transformers library. The model generated a 768-dimensional dense vector for each document using mean pooling over the final hidden state. These embeddings allowed us to measure the contextual similarity between a user query and candidate documents using cosine distance.

## 4.3 Model Training and Hybrid Search Logic

The hybrid model was designed to leverage both TF-IDF and BERT in a two-step ranking process:

- Step 1 – Lexical Filtering with TF-IDF: Given a user query, we computed its TF-IDF vector and retrieved the top-N (typically 2×k) most similar documents based on cosine similarity.
- Step 2 – Semantic Re-Ranking with BERT: The filtered documents were then encoded using BERT, along with the input query. We computed cosine similarity between the query embedding and document embeddings to assess semantic relevance.
- Final Scoring: The overall score for each document was computed as a weighted combination of the TF-IDF score and the BERT score:
- Final_Score = α × TFIDF_Score + (1 - α) × BERT_Score
- The α parameter (typically 0.4 to 0.5) allowed us to balance lexical and semantic contributions to the final ranking.

## 4.4 Performance Evaluation

We evaluated the model using a combination of qualitative assessments and response-time benchmarks. Test queries were run in three configurations: TF-IDF only, BERT only, and the hybrid pipeline. Results were reviewed manually to assess semantic alignment with the queries. The hybrid model consistently retrieved more relevant documents than TF-IDF alone, and did so more efficiently than a full BERT-based search.

Average query latency was significantly reduced due to the TF-IDF filtering step, making the system scalable for larger document sets. This methodology demonstrates how hybrid search models can balance accuracy and efficiency in real-world document retrieval applications.

## 4.3 Model Training and Evaluation

The final phase of our methodology focused on building and evaluating a hybrid document retrieval model that integrates traditional keyword-based search with deep contextual understanding. The model leverages both TF-IDF and BERT to rank documents in response to user queries, optimizing for accuracy, relevance, and scalability.

TF-IDF Filtering

We began by training a TfidfVectorizer on the full corpus of cleaned document text. This vectorizer transforms each document into a sparse numerical representation, where each dimension corresponds to a unique word weighted by its importance. For a given input query, TF-IDF scores are computed using cosine similarity, allowing us to efficiently shortlist a subset of the most lexically similar documents.

BERT Embedding & Semantic Ranking

Once filtered, the top-N documents (typically twice the number of desired results) are passed to a pre-trained BERT model ("bert-base-uncased") to compute semantic embeddings. Both the query and candidate documents are encoded using mean-pooled representations of their final hidden states. Cosine similarity between the query and each document embedding provides a semantic relevance score.

Hybrid Score Computation

The final document ranking is determined by a weighted combination of TF-IDF and BERT scores:

$$\text{Final_Score} = \alpha \times \text{TFIDF_Score} + (1 - \alpha) \times \text{BERT_Score}$$

Where $\alpha$ is a tunable parameter (typically set to 0.4–0.5). This allows balancing between surface-level keyword relevance and deeper semantic similarity, giving the user more accurate results with reduced computational cost.

Evaluation Strategy

Given the subjective nature of relevance in document retrieval, we adopted both qualitative and semi-quantitative evaluation approaches:

- Top-K Accuracy: We manually evaluated the relevance of top-5 results for a diverse set of queries. The hybrid model consistently outperformed TF-IDF-only and BERT-only approaches in balancing precision and computational efficiency.
- Inference Time: Runtime benchmarks showed a 3–5× reduction in average query latency compared to using BERT on the entire document set.
- User Relevance Feedback: Example queries demonstrated that hybrid search returned semantically meaningful documents even when query keywords were absent from the result text.

The evaluation confirms that our hybrid model offers significant performance improvements for real-world document search tasks. By narrowing the candidate set using TF-IDF and applying BERT selectively, we achieve a practical trade-off between speed and semantic depth.

## 4.3 Model Training and Evaluation

To assess the effectiveness of our hybrid document retrieval system, we conducted a multi-dimensional performance evaluation. The core goal was to determine how well the combination of TF-IDF and BERT balances speed, relevance, and contextual understanding when compared to using either technique independently.

We focused on four key evaluation aspects: (1) Top-K Relevance, which measures how contextually relevant the top retrieved documents are to the input query; (2) Precision at K, which considers the fraction of retrieved results that are semantically meaningful; (3) Inference Time, to assess the average latency per query; and (4) Subjective User Quality, based on human assessment of search outputs.

To conduct this evaluation, we created a test suite of fifteen diverse user queries across domains such as law, education, healthcare, and technology. Each query was executed using three configurations: TF-IDF-only filtering, BERT-only semantic ranking, and the combined hybrid approach. Returned results were reviewed and scored by multiple evaluators based on how relevant they were to the query intent.

The results showed that TF-IDF alone was fast but often returned documents that were only partially relevant, especially for queries involving abstract or indirect language. BERT-only retrieval produced highly accurate results due to its deep semantic understanding but was significantly slower, as it required computing embeddings for the entire corpus.

The hybrid approach delivered the best of both worlds. It filtered the candidate documents using TF-IDF, reducing the search space, and then re-ranked this subset using BERT embeddings. This process not only improved semantic accuracy but also reduced inference time substantially compared to a full BERT-only pipeline.

For example, in a legal-domain query about environmental education policies, the hybrid model retrieved documents that discussed constitutional clauses and environmental law initiatives—even though they didn't share exact keyword matches with the query. TF-IDF alone missed these documents, while BERT-only retrieval succeeded but at a much higher computational cost. The hybrid approach surfaced these relevant documents quickly and accurately.

Overall, the evaluation confirmed that our hybrid document search engine is highly effective for real-world use. It balances performance and precision, scales well with document volume, and provides a solid foundation for intelligent, context-aware document retrieval.

## 5.2 Feature Importance Analysis

To gain interpretability and transparency into the model's decision-making process, we conducted a detailed feature importance analysis. Understanding which features contribute most to the model's predictions helps validate domain assumptions and guides further feature engineering efforts.
Since the hybrid search system integrates both traditional vector-space representations (TF-IDF) and dense embeddings (BERT), we evaluated feature influence in both contexts.

**TF-IDF-Based Importance**

In the TF-IDF component, feature importance was derived from term frequency and inverse document frequency weights. Words that appeared frequently in relevant documents but were rare across the entire corpus received higher weights and were more likely to influence initial document selection. For instance, domain-specific terms such as "constitutional amendment," "school policy," or "loan default" ranked among the most discriminative keywords in their respective query spaces. This indicates that TF-IDF served as a useful lexical filter by prioritizing documents containing rare yet relevant terms.

**Semantic Features via BERT**

For the BERT-based ranking stage, feature importance is distributed across the 768-dimensional contextual embedding generated for each document and query. Although individual embedding dimensions are not easily interpretable, we used cosine similarity to measure the alignment between the query vector and document vectors. To analyze this semantically, we examined the text spans that contributed most to the BERT similarity scores.
Using attention heatmaps and token saliency techniques, we found that BERT consistently focused on contextually meaningful phrases, even when they were lexically dissimilar to the query. For example, for a query on "educational reforms in environment law," BERT emphasized semantically aligned content such as "eco-friendly curriculum" and "green policy advocacy," even when these phrases didn't share literal keywords with the query. This underscores BERT's ability to recognize conceptual relevance beyond surface-level matches.

**Comparative Insights**

Overall, TF-IDF contributed primarily to high-precision keyword filtering, while BERT enabled high-recall semantic matching. Their combined importance analysis revealed that surface-level tokens acted as gatekeepers, while deeper context—captured through semantic embeddings—was critical for final ranking decisions.
This layered importance structure affirms the value of a hybrid system where lexical specificity is used to narrow down candidates, and semantic understanding is used to refine the final results. Such a design not only improves performance but also supports explainability, which is especially valuable in sensitive domains like legal or academic search systems.

# 6. Conclusion

This project set out to develop a hybrid document retrieval system that combines the speed of TF-IDF-based lexical filtering with the semantic depth of BERT embeddings. By integrating these two powerful techniques, we successfully addressed the limitations of traditional keyword search while avoiding the high computational cost of deep learning applied at scale.

Our methodology involved thorough data preprocessing, feature engineering, and vector representation, followed by the implementation of a two-stage retrieval pipeline. TF-IDF served as an efficient first-pass filter to reduce the document pool, while BERT was used to semantically re-rank the top candidates. This hybrid approach allowed us to capture both explicit keyword matches and contextual relevance, providing users with more meaningful and accurate search results.

Performance evaluations demonstrated that the hybrid model consistently outperformed standalone TF-IDF and BERT configurations in both retrieval quality and inference speed. Feature importance analysis further validated the model's ability to distinguish useful patterns, and exploratory insights such as the impact of business size on default risk reinforced the model's real-world applicability.

In summary, the hybrid search engine presented in this project offers a scalable, intelligent, and practical solution for modern document discovery tasks. Its flexibility makes it well-suited for applications in legal research, academic indexing, corporate knowledge systems, and more.

## 7. Future Work

While the current hybrid document retrieval system achieves a practical balance between accuracy and efficiency, several enhancements can be explored to further improve its performance, scalability, and real-world applicability.

1. Domain-Specific Fine-Tuning
2. The current implementation uses a general-purpose BERT model (bert-base-uncased). Fine-tuning the model on domain-specific corpora—such as legal, academic, or financial texts—could significantly improve semantic understanding and retrieval relevance in specialized applications.
3. Embedding Caching and Indexing
4. Currently, BERT embeddings are computed on-the-fly for filtered documents. To reduce latency in large-scale deployments, we can precompute and cache document embeddings. Integration with vector indexing tools like FAISS or Annoy would enable high-speed similarity search, making the system more suitable for real-time environments.
5. UI/UX and Web Deployment
6. Building an interactive web-based interface would allow end users to query the system more intuitively. A front-end portal integrated with a backend API could support features like query autosuggestions, semantic filters, and result summarization for better usability.
7. Result Summarization and Snippet Highlighting
8. To further improve the user experience, implementing a summarization module could generate concise previews of the most relevant sections in each document. Highlighting matching phrases from both TF-IDF and BERT perspectives would help users understand why a result was returned.
9. Multilingual and Cross-Lingual Search
10. The current model is limited to English text. Extending the system with multilingual support using models like XLM-RoBERTa would enable retrieval across languages, which is especially beneficial for international legal or academic corpora.
11. Query Expansion and Reformulation
12. Incorporating query expansion techniques—such as synonym retrieval, entity linking, or BERT-based paraphrasing—could improve recall and ensure that relevant documents are not missed due to vocabulary mismatch.
13. Relevance Feedback and Active Learning
14. Integrating user feedback loops to capture clicks or ratings could enable the model to adapt over time. Active learning techniques could also help retrain ranking mechanisms using real-world query-response pairs.

These future directions present valuable opportunities to transform the current prototype into a robust, scalable, and intelligent semantic search platform capable of supporting a wide range of knowledge retrieval applications.

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